

Junho Peter Whang

Department of Mathematical Sciences
Seoul National University

jhpwhang@gmail.com

jhpwhang.github.io

Citizenship: Canada

Academic Positions

Seoul National University, Seoul, Korea

Aug. 2021–present

Assistant Professor, Department of Mathematical Sciences

Princeton University, Princeton, NJ, USA

Sep. 2025–Jun. 2026

Visiting Fellow, Department of Mathematics (sabbatical)

Massachusetts Institute of Technology, Cambridge, MA, USA

Jul. 2018–Jun. 2021

C.L.E. Moore Instructor

Education

Princeton University, Princeton, NJ, USA

2018

Ph.D. in Mathematics. Advisor: Peter Sarnak.

Thesis: *Diophantine Analysis on Moduli of Local Systems*.

Queen's University, Kingston, ON, Canada

2013

B.Sc. in Mathematics, Honours with Distinction.

Research Interests

Number theory (Diophantine analysis), arithmetic geometry, topology, and dynamics.

Publications and Preprints

1. *On indefinite integral ternary quadratic forms*.
With Alexander Gamburd, Amit Ghosh, and Peter Sarnak. Preprint, arXiv:2603.05849.
2. *Degree gaps in algebraic interpolations*.
Bull. Korean Math. Soc. **63** (2026), no. 1, 227–234.
3. *A reciprocity law on the one-holed torus*.
J. Korean Math. Soc. **62** (2025), no. 6, 1525–1538.
4. *On finite categories of algebraic varieties*.
J. Algebra **681** (2025), 206–217.
5. *Stokes matrices and exceptional isomorphisms*.
With Yu-Wei Fan. Math. Ann. **390** (2024), 4041–4086.
6. *On periodic orbits of polynomial maps*.
Preprint, arXiv:2305.13529.
7. *Surface group representations in $\mathrm{SL}_2(\mathbb{C})$ with finite mapping class orbits*.
With Indranil Biswas, Subhojoy Gupta, and Mahan Mj. Geom. Topol. **26** (2022), no. 2, 679–719.
8. *The rank two p -curvature conjecture on generic curves*.
With Anand Patel and Ananth Shankar. Adv. Math. **386** (2021).

9. *Arithmetic of curves on moduli of local systems.*
Algebra Number Theory **14** (2020), no. 10, 2575–2605.
10. *Nonlinear descent on moduli of local systems.*
Israel J. Math. **240** (2020), 935–1004.
11. *Global geometry on moduli of local systems for surfaces with boundary.*
Compos. Math. **156** (2020), no. 8, 1517–1559.
12. *The uncertainty principle and a generalization of a theorem of Tao.*
With M. Ram Murty. Linear Algebra Appl. **437** (2012), no. 1, 214–220.
13. *Another proof of the infinitude of the prime numbers.*
Amer. Math. Monthly **117** (2010), no. 2, 181.

Grants and Research Funding

Samsung Science and Technology Foundation	2022–2027
<i>Analysis of Diophantine Problems with Nonlinear Symmetry.</i> SSTF-BA2201-03.	
Principal Investigator. Total project award: KRW 755 million.	
National Research Foundation of Korea	2025–2028
<i>Dynamics and Arithmetic Structures on Manifolds (DASOM).</i> RS-2025-02293115.	
Co-investigator. Total project award: KRW 500 million.	

Selected Honors and Awards

Teaching and Learning Award , Massachusetts Institute of Technology	2021
Bershadsky Family Fellowship in Mathematics , Princeton University	2017–2018
NSERC Postgraduate Scholarship PGS D	2014–2017
NSERC Postgraduate Scholarship PGS M	2013–2014
Governor General’s Academic Medal , Queen’s University	2013
Prince of Wales Prize , Queen’s University	2013

Teaching

Seoul National University	2021–present
<i>Responsibilities:</i> Course design and delivery, preparation of assignments and examinations, assessment, and student advising through office hours.	
<i>Graduate courses</i>	
• Algebra III	Spring 2025
• Algebra II	Fall 2024
• Algebra I	Spring 2024
• Commutative Algebra	Spring 2022, Spring 2024
• Algebraic Geometry	Spring 2023

Graduate topics courses

- Diophantine Analysis Fall 2026 (current)
- Discrete Groups and Diophantine Problems Spring 2025
- Schemes: Cohomology and Applications Fall 2023
- Diophantine Geometry Spring 2023

Undergraduate courses

- Analytic Number Theory (topics course) Fall 2026 (current)
- Linear Algebra Fall 2022
- Introduction to Diophantine Analysis (topics course) Fall 2021

Massachusetts Institute of Technology

2018–2021

- Instructor: Introduction to Arithmetic Geometry (18.782), Fall 2019.
- Course administrator and recitation leader: Multivariable Calculus (18.02) and Differential Equations (18.03), various semesters.

Research Supervision

Ph.D. students

- Seong Youn Kim, 2022–present; co-advised with Gye-Seon Lee.
- Young Joon Ley, 2022–present.
- Eungchan Kim, 2025–present.
- Jihye Oh, 2025–present.

Master’s students

- Juhan Kim, M.S. 2023.
Thesis: *Compactification of Character Varieties via Triangulation on Surfaces.*
- Eunju Shin, M.S. 2024.
Thesis: *Nonlinear Reduction for Two Types of Generalized Markoff Equations.*
- Jihye Oh, M.S. 2025.
Thesis: *Decidability of Mutation Equivalence for Three-Vertex Quivers.*

Last updated: September 16, 2026